

5G RAN

mmWave Super Coverage (High-Frequency TDD) Feature Parameter Description

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1 Change History

The following describes the specific changes in different issues.

5G RAN10.1 Draft B (2026-01-30)

This issue includes the following changes.

- Technical changes
None
- Editorial changes
Revised descriptions in this document, which do not involve changes in meanings and therefore are not detailed.

5G RAN10.1 Draft A (2025-12-31)

This issue is the first release.

2 About This Document

2.1 General Statements

Purpose

Feature Parameter Description documents are intended to acquaint readers with:

- The technical principles of features and their related parameters
- The scenarios where these features are used, the benefits they provide, and the impact they have on networks and functions
- Requirements that must be met before feature activation
- Parameter configuration required for feature activation, verification of feature activation, and monitoring of feature performance

NOTE

This document only provides guidance for feature activation. Feature deployment and feature gains depend on the specifics of the network scenario where the feature is deployed. To achieve optimal gains, contact Huawei professional service engineers.

Functions mentioned in this document work properly only when enabled in the specified applicable scenarios (such as RAT and networking). If a function not mentioned in this document is enabled or a function is enabled in a scenario not specified as applicable, exceptions or other impacts may occur.

Software Interfaces

Any parameters, alarms, counters, or managed objects (MOs) described in Feature Parameter Description documents apply only to the corresponding software release. For future software releases, refer to the corresponding updated product documentation.

2.2 Features in This Document

This document describes the following features.

Feature ID	Feature Name	Chapter/Section
FOFD-101242	mmWave Super Coverage	4 mmWave Super Coverage

2.3 Differences

Differences Between NSA and SA

Function Name	Difference	Chapter/Section
Intelligent beam selection	None	4.1.1 Intelligent Beam Selection
Beam tracking	None	4.1.2 Beam Tracking
SRS-based digital weight	None	4.1.3 SRS-based Digital Weight
SRS bandwidth adaptation	None	4.1.4 SRS Bandwidth Adaptation
Precise power control	None	4.1.5 Precise Power Control
Downlink IBLER adaptation for CEUs	None	4.1.6 Downlink IBLER Adaptation for CEUs

Differences Between High Frequency Bands and Low Frequency Bands

This document refers to frequency bands belonging to FR1 (410–7125 MHz) as low frequency bands, and those belonging to FR2 (24250–71000 MHz) as high frequency bands. For details about FR1 and FR2, see section 5.1 "General" in 3GPP TS 38.104 V17.8.0.

Function Name	Difference	Chapter/Section
Intelligent beam selection	Supported only in high frequency bands	4.1.1 Intelligent Beam Selection
Beam tracking	Supported only in high frequency bands	4.1.2 Beam Tracking

Function Name	Difference	Chapter/Section
SRS-based digital weight	Fixed SRS-based digital weight and SRS- and PMI-based weight adaptation: This document describes these functions only in high-frequency scenarios. For details about these functions in low-frequency scenarios, see <i>MIMO (TDD)</i>. SRS filtering-based downlink beamforming weight optimization: This document describes this function only in high-frequency scenarios. For details about this function in low-frequency scenarios, see <i>FWA</i>. Other functions: supported only in high frequency bands	4.1.3 SRS-based Digital Weight
SRS bandwidth adaptation	Supported only in high frequency bands	4.1.4 SRS Bandwidth Adaptation
Precise power control	PDSCH DMRS power adaptation: This document describes this function only in high-frequency scenarios. For details about this function in low-frequency scenarios, see <i>Power Control</i>. Other functions: supported only in high frequency bands	4.1.5 Precise Power Control
Downlink IBLER adaptation for CEUs	Supported only in high frequency bands	4.1.6 Downlink IBLER Adaptation for CEUs

3 Overview

The mmWave super coverage function is designed to cope with the weaknesses of mmWave bands in an NR system—such as weaker coverage than low frequency bands due to high propagation loss and susceptibility to obstacles. This feature improves downlink cell coverage and user experience using beam directivity accuracy enhancement, adaptive optimization, channel noise reduction, channel prediction, and spectral efficiency optimization. This feature applies to high-frequency TDD cells with coverage requirements or scenarios with user experience improvement requirements.

4 mmWave Super Coverage

4.1 Principles

mmWave super coverage is controlled by the **SUPER_COVERAGE_SW** option of the **NRDUCellChnCovAlgo.DICoverageAlgoSwitch** parameter and includes the functions listed in [Table 4-1](#).

Table 4-1 Functions of mmWave Super Coverage

Function Name	Description	Target UE	Chapter/Section
Intelligent beam selection	This function is an enhancement to Flexible Dense Beam. After Flexible Dense Beam is enabled, intelligent beam selection helps the gNodeB to more accurately direct CSI-RS for BM beams at UEs by increasing the number of CSI-RS for BM beams and applying the neural network technology. This increases the average uplink and downlink UE throughputs.	UEs with uplink large-packet services and UEs performing downlink large-packet services	4.1.1 Intelligent Beam Selection
Beam tracking	This function enables the gNodeB to predict the UE trajectories, and select the best CSI-RS for BM beam as the serving beam for these UEs. This increases the average uplink and downlink UE throughputs in mobility scenarios.	All UEs	4.1.2 Beam Tracking

Function Name	Description	Target UE	Chapter/Section
SRS-based digital weight	This function enables a base station to transmit beams in multiple directions. As a result, the beams better match channel conditions, thereby increasing the average downlink UE throughput.	All UEs	4.1.3 SRS-based Digital Weight
SRS bandwidth adaptation	This function enables adaptive switching of SRS transmission mode for UEs based on SRS channel quality measurements. This improves the SRS quality and enhances the accuracy of SRS-based weights, thereby increasing the average downlink throughput of non-cell-center UEs.	Non-cell-center UEs performing downlink large-packet services	4.1.4 SRS Bandwidth Adaptation
Precise power control	This function enables the gNodeB to aggregate power resources in the frequency domain for cell-edge UEs (CEUs) to increase downlink spectral efficiency, thereby increasing the average downlink throughput of CEUs.	Inter-FR CA CEUs using extremely low MCS indexes	4.1.5 Precise Power Control
Downlink IBLER adaptation for CEUs	This function enables the gNodeB to adaptively select the target IBLER based on channel fluctuation, thereby increasing the average downlink throughput of CEUs.	CEUs using extremely low MCS indexes and experiencing significant channel condition fluctuations	4.1.6 Downlink IBLER Adaptation for CEUs

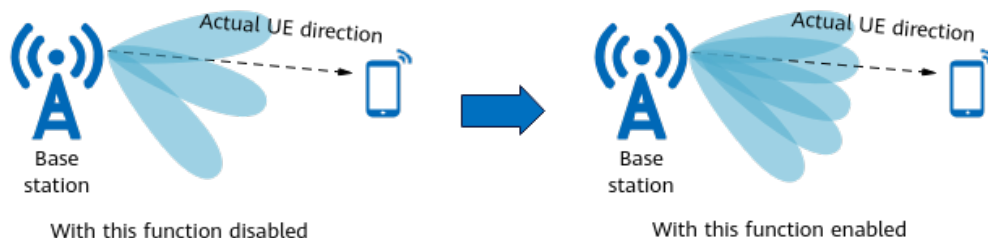
4.1.1 Intelligent Beam Selection

Flexible Dense Beam strengthens basic CSI-RS for BM beams, thereby mitigating beam dents. However, Flexible Dense Beam alone may not sufficiently mitigate beam dents, if CSI-RS for BM beams become narrower due to an RF module containing a larger number of antenna elements. Against this backdrop, due to the limited beam sweeping capability of UEs, Flexible Dense Beam is insufficient for further densifying beams. To address this issue, intelligent beam selection is introduced. For details about Flexible Dense Beam, see *mmWave Beam Management (High-Frequency TDD)*.

As an enhancement to Flexible Dense Beam, intelligent beam selection uses the neural network technology and increases the number of CSI-RS for BM beams to

help the gNodeB accurately direct CSI-RS for BM beams at UEs, thereby increasing average uplink and downlink UE throughputs. For details, see [Figure 4-1](#).

Figure 4-1 Intelligent beam selection



Specifically, intelligent beam selection works as follows: (1) The gNodeB densifies the CSI-RS for BM beams formed by Flexible Dense Beam to obtain a set of denser candidate CSI-RS for BM beams. (2) A UE measures these candidate beams. (3) Based on the UE-reported beam measurements, the gNodeB uses a neural network model to select CSI-RS for BM beams that more accurately direct at UEs from the candidate beams and then select the best beam to serve the UE.

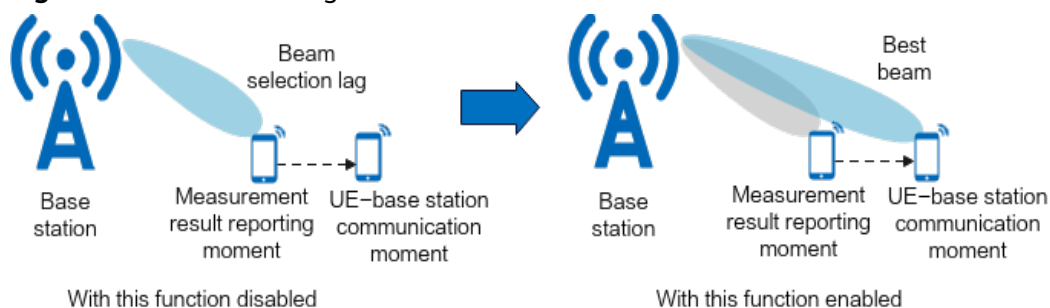
Intelligent beam selection is controlled by the **INTELLIGENT_BEAM_SELECTION_SW** option of the **NRDUCellBeamAlgo.BeamOptAlgoSwitch** parameter.

Intelligent beam selection is applicable only to UEs performing uplink or downlink large-packet services in a cell.

4.1.2 Beam Tracking

In an mmWave communications system, the gNodeB selects the best beam as the serving beam for a UE based on the CSI-RS for BM beam measurements reported by the UE. For a moving UE in a cell, beam selection by the gNodeB lags behind. A higher moving speed and a longer reporting period of the beam measurement result lead to greater impacts of the lag on communications performance. In this context, beam tracking is introduced to predict the trajectory of the moving UE and select the best CSI-RS for BM beam as the serving beam, thereby improving the average uplink and downlink UE throughputs in mobility scenarios. For details, see [Figure 4-2](#).

Figure 4-2 Beam tracking



After beam tracking is enabled, the gNodeB predicts the trajectory of a moving UE using a reinforcement learning model and CSI-RS for BM beam measurements.

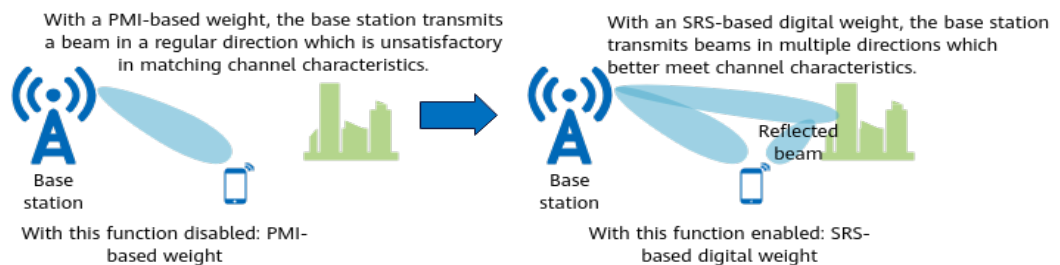
Based on the predicted trajectory, the gNodeB then predicts the best beam to serve the UE for the next moment. The gNodeB finally selects the best CSI-RS for BM beam as the serving beam for the UE, based on the prediction and UE-reported measurements.

Beam tracking is controlled by the **BEAM_TRACKING_SW** option of the **NRDUCellBeamAlgo.BeamOptAlgoSwitch** parameter.

4.1.3 SRS-based Digital Weight

In NR TDD, the uplink and downlink channels over the air interface are reciprocal. A PMI-based weight is the weight best suitable for the current channel status, which is selected from the codebook based on the feedback from UEs in a cell and used for channel weighting. A PMI-based weight is configured by deselecting the **DL_PMI_SRS_ADAPT_SW** option of the **NRDUCellAlgoSwitch.AdaptiveEdgeExpEnhSwitch** parameter and setting the **NRDUCellPdsch.FixedWeightType** parameter to **PMI_WEIGHT**. An SRS-based digital weight is calculated based on the uplink channel measurement result and used for downlink channel weighting. This makes the SRS-based digital weight better match channel characteristics. Compared with a PMI-based weight, an SRS-based digital weight has a finer granularity in frequency-domain weighting, which reduces the throughput loss caused by frequency selection and thereby increases the average downlink UE throughput. [Figure 4-3](#) shows the working principles of SRS-based digital weight.

Figure 4-3 SRS-based digital weight



SRS-based digital weight supports fixed SRS-based digital weight and SRS- and PMI-based weight adaptation:

- Fixed SRS-based digital weight: This function is enabled when the **DL_PMI_SRS_ADAPT_SW** option of the **NRDUCellAlgoSwitch.AdaptiveEdgeExpEnhSwitch** parameter is deselected and the **NRDUCellPdsch.FixedWeightType** parameter is set to **SRS_WEIGHT**. The gNodeB uses a fixed SRS-based weight.
- SRS- and PMI-based weight adaptation: This function is enabled when the **DL_PMI_SRS_ADAPT_SW** option of the **NRDUCellAlgoSwitch.AdaptiveEdgeExpEnhSwitch** parameter is selected. The gNodeB adaptively selects the SRS-based digital weight or PMI-based weight based on the spectral efficiency.

SRS-based digital weight also incorporates optimization functions:

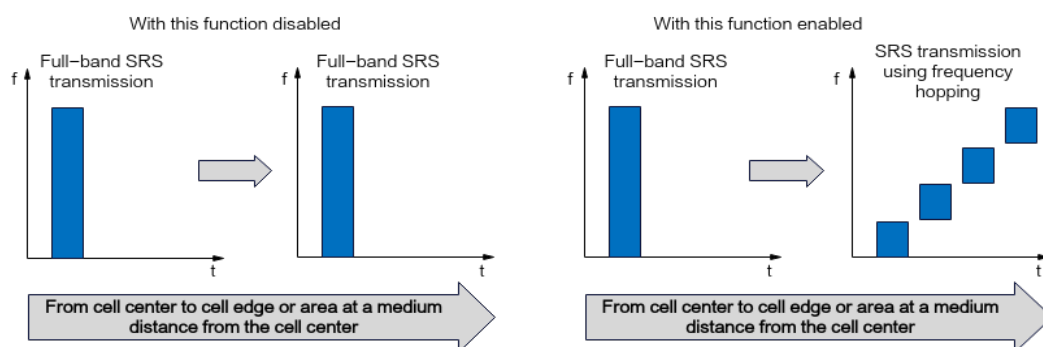
- Downlink single-UE mobility precoding: This function is enabled when the **NRDUCellPdschPrecode.DLSuMimoMobPrecodeMode** parameter is set to **ROBUST_PHASEIII**. With this function, a prediction model is trained through a neural network and then used to predict channels, thereby alleviating the loss caused by channel changes and increasing the average downlink UE throughput. This function takes effect only for UEs performing downlink large-packet services.
- SRS filtering-based downlink beamforming weight optimization: This function is enabled when the **DL_BF_WEIGHTFILTER_OPT_SW** option of the **NRDUCellAlgoSwitch.AdaptiveEdgeExpEnhSwitch** parameter is selected. In FWA scenarios, this function implements smooth noise reduction on channels at several moments, thereby increasing the SNR and average downlink UE throughput.

4.1.4 SRS Bandwidth Adaptation

Assume that a UE has a low SRS SNR. For example, a non-cell-center UE typically has a low SRS SNR. For such a UE, either the SRS-based digital weight does not take effect, or its estimated value is inaccurate and misaligned with the actual downlink data channel conditions. This results in poor performance of the non-cell-center UE. Against this backdrop, SRS bandwidth adaptation is introduced.

After this function is enabled, the gNodeB controls the adaptive switching of SRS transmission modes based on the SRS-based channel quality measurement result. SRS frequency-hopping transmission can be used to increase the SRS SNR and proportion of cases that the SRS-based digital weight takes effect. In this way, the SRS quality is improved and the accuracy of the SRS-based weight is enhanced, thereby increasing the average downlink throughput of the non-cell-center UE. For details, see [Figure 4-4](#). In this process, SRS frequency hopping enables SRS transmission on a small bandwidth in a symbol and on a different bandwidth in the next symbol. The same rule applies to the rest SRSs. In this way, the gNodeB achieves sounding on the entire channel bandwidth. Frequency hopping offers channel quality measurement across the entire bandwidth using a small sounding bandwidth.

Figure 4-4 SRS bandwidth adaptation



SRS bandwidth adaptation is controlled by the **SRS_BANDWIDTH_ADAPT_SW** option of the **NRDUCellSrs.SrsAlgoExtSwitch** parameter.

SRS bandwidth adaptation takes effect only for one UE performing downlink large-packet services in a sector at a time.

4.1.5 Precise Power Control

Precise power control is designed to enhance the coverage of inter-FR CA CEUs using extremely low MCS indexes. This function dynamically adjusts the power spectral density (PSD) of downlink data channels (involving downlink data and DMRS) to improve downlink spectral efficiency and demodulation performance, thereby improving the average downlink throughput of CEUs. Precise power control includes two sub-functions: downlink PDSCH coverage enhancement and PDSCH DMRS power adaptation. It is recommended that both sub-functions be enabled.

- Downlink PDSCH coverage enhancement
This function is controlled by the **PDSCH_COV_ENHANCE_SW** option of the **NRDUCellChnPwr.ChnPwrAlgoSwitch** parameter. With this function, the PDSCH PSD can be adaptively adjusted based on the MCS indexes and the number of RBs at the cell reference power to support higher MCS indexes. This improves the spectral efficiency of UEs, thereby increasing the downlink cell coverage and average downlink UE throughput. For details about the cell reference power, see *Power Control*.
- PDSCH DMRS power adaptation
This function is controlled by the **DMRS_PWR_ADAPT_SW** option of the **NRDUCellChnPwr.ChnPwrAlgoSwitch** parameter. With this function, the transmission power of data which is on the same symbol with the PDSCH DMRS is adaptively allocated to the PDSCH DMRS. This improves the demodulation performance of CEUs, thereby enabling their support for higher MCS indexes. As a result, the spectral efficiency of UEs is improved, thereby enhancing the downlink cell coverage and average downlink UE throughput.

NOTE

- Extremely low MCS indexes refer to MCS 0 in the DL 256QAM MCS index table and MCS 0, MCS 1, and MCS 2 in the DL 64QAM MCS index table. For details about the DL 256QAM MCS index table and DL 64QAM MCS index table, see section 5.1.3.1 "Modulation order and target code rate determination" of 3GPP TS 38.214 V17.1.0.
- For details about inter-FR CA, see *Carrier Aggregation*.

4.1.6 Downlink IBLER Adaptation for CEUs

Downlink IBLER adaptation for CEUs is designed to enhance the coverage of CEUs using extremely low MCS indexes and experiencing significant channel condition fluctuations. This function is controlled by the **DL_EDGE_IBLER_ADAPT_SW** option of the **NRDUCellDLAmc.DIMcsSelAlgoSw** parameter. With this function, the gNodeB adaptively selects the downlink target IBLER based on (1) channel condition fluctuations and (2) whether CEUs use extremely low MCS indexes. This improves the average MCS index for downlink scheduling, thereby increasing the spectral efficiency and average downlink throughput of CEUs.

When downlink IBLER adaptation for CEUs does not take effect, the downlink target IBLER for UEs is fixed at a value specified by the **NRDUCellPdsch.DITargetIbler** parameter.

 NOTE

Average MCS index for downlink scheduling = $(0 \times \text{N.ChMeas.PDSCH.MCS.0} + 1 \times \text{N.ChMeas.PDSCH.MCS.1} + 2 \times \text{N.ChMeas.PDSCH.MCS.2} + 3 \times \text{N.ChMeas.PDSCH.MCS.3} + \dots + 26 \times \text{N.ChMeas.PDSCH.MCS.26} + 27 \times \text{N.ChMeas.PDSCH.MCS.27} + 28 \times \text{N.ChMeas.PDSCH.MCS.28}) / (\text{N.ChMeas.PDSCH.MCS.0} + \text{N.ChMeas.PDSCH.MCS.1} + \text{N.ChMeas.PDSCH.MCS.2} + \text{N.ChMeas.PDSCH.MCS.3} + \dots + \text{N.ChMeas.PDSCH.MCS.26} + \text{N.ChMeas.PDSCH.MCS.27} + \text{N.ChMeas.PDSCH.MCS.28})$

4.2 Network Analysis

4.2.1 Benefits

[Table 4-2](#) describes the benefits of each function after mmWave super coverage is enabled.

Table 4-2 Benefits

Function Name	Benefit
Intelligent beam selection	- After intelligent beam selection is enabled, the values of User Downlink Average Throughput (DU) and User Uplink Average Throughput (DU) increase for the UEs with this function in effect in the cell. The benefits are related to the locations of UEs with this function in effect: - There are greater benefits when UEs with this function in effect are closer to the dents of Flexible Dense Beams. - There are less or even no benefits when UEs with this function in effect are closer to the wave peak of Flexible Dense Beams. - After intelligent beam selection is enabled, the gNodeB collects data and trains the neural network model. During this period, no benefits are provided. When the status of the mmWave Super Coverage or intelligent beam selection switch changes from on to off, the neural network model's related data is cleared. In this case, if the switch is turned on again, the gNodeB collects data and trains the neural network model again. During this period, no benefits are provided. Benefit monitoring is recommended when the neural network model is in the In Use state. Run the DSP NRDUCELLBMAIMODEL command with the NRDUCellBmAiModel.AiCase parameter set to INTELLIGENT_BEAM_SELECTION. Observe the value of AI Model Status. The value In Use indicates that the neural network model is in use. NOTE After intelligent beam selection is enabled, run the DSP NRDUCELLBMAIMODEL command with the NRDUCellBmAiModel.AiCase parameter set to INTELLIGENT_BEAM_SELECTION. Observe the value of AI Model Status. The possible values include Building, In Use, and Updating. Intelligent beam selection provides benefits only when the value is In Use. When the neural network model is in use, the gNodeB monitors the model performance in real time. If the gNodeB detects that the model parameters are aged, it automatically collects data and trains the neural network model again. During this process, the value of AI Model Status is Updating, and no benefits are provided.

Function Name	Benefit
Beam tracking	- After beam tracking is enabled, the average uplink and downlink throughputs increase for the UEs with this function in effect in the cell. This function is recommended in eMBB scenarios. The benefits are related to the following factors: - A higher UE moving speed results in a longer beam selection lag and greater benefits. A lower UE moving speed results in a shorter beam selection lag and less benefits. - A higher proportion of LOS UEs results in a higher accuracy rate of UE trajectory prediction and greater benefits. A higher proportion of NLOS UEs results in a lower accuracy rate of UE trajectory prediction and less benefits. - After beam tracking is enabled, the gNodeB collects data and trains the reinforcement learning model. During this period, less or even no benefits are provided. When the status of the mmWave Super Coverage or beam tracking switch changes from on to off, the reinforcement learning model's related data is cleared. In this case, if the switch is turned on again, the gNodeB collects data and trains the reinforcement learning model again. During this period, less or even no benefits are provided. Benefit monitoring is recommended when the reinforcement learning model is in the In Use state. Run the DSP NRDUCELLBMAIMODEL command with the NRDUCellBmAiModel.AiCase parameter set to BEAM_TRACKING. Observe the value of AI Model Status. The value In Use indicates that the reinforcement learning model is in use.

Function Name	Benefit
SRS-based digital weight	- Fixed SRS-based digital weight: The average downlink UE throughput increases. A higher proportion of NLOS scenarios results in greater benefits. A higher proportion of LOS scenarios results in fewer or even no benefits. - SRS- and PMI-based weight adaptation: The average downlink UE throughput increases. The benefits are related to the following factors: - A higher UE moving speed results in faster channel changes, lower weight correlation, and less benefits. - A larger number of UEs results in a longer interval of aperiodic SRS reporting, and a longer weight update interval. As a result, the weight correlation decreases, causing reduced benefits. - Downlink single-UE mobility precoding: The gNodeB predicts channel changes to alleviate the loss caused by channel changes, thereby increasing the average downlink UE throughput. This function is recommended in eMBB scenarios. The benefits are related to the following factors: - A longer SRS period results in lower SRS channel correlation and higher benefits from downlink single-UE mobility precoding. - A larger SRS bandwidth results in more frequent channel sampling, higher accuracy of "downlink single-UE mobility precoding", and greater benefits. - A higher UE moving speed results in faster channel changes, lower SRS channel correlation, and higher benefit potential from downlink single-UE mobility precoding. However, an extremely high moving speed leads to overly fast channel changes and thereby lower SRS channel correlation. As a result, accurate channel prediction cannot be realized even through "downlink single-UE mobility precoding", thereby reducing the benefits or even causing no benefits. A lower UE moving speed results in slower channel changes, higher SRS channel correlation, and lower benefit potential or even no benefits from downlink single-UE mobility precoding. - A larger number of antennas in a cell results in denser channel sampling in the space domain, higher accuracy of "downlink single-UE mobility precoding", and greater benefits. - In NLOS scenarios, the aging of SRS-based digital weights occurs, resulting in a high benefit potential of "downlink single-UE mobility precoding". In LOS scenarios, SRS-based digital weights do not age, resulting in a low benefit potential or even no benefits of "downlink single-UE mobility precoding".

Function Name	Benefit
	• SRS filtering-based downlink beamforming weight optimization: The gNodeB performs noise reduction on channels to increase the downlink UE throughput. This function is recommended in FWA scenarios. The benefits are related to the following factors: – When the SNR of a UE is less than or equal to -20 dB, "SRS filtering-based downlink beamforming weight optimization" does not produce benefits. – When the SNR of a UE is greater than -20 dB and less than 10 dB, "SRS filtering-based downlink beamforming weight optimization" has obvious noise reduction effect and produces high benefits. – When the SNR of a UE is greater than or equal to 10 dB, the impact of noise is small and "SRS filtering-based downlink beamforming weight optimization" produces low benefits.
SRS bandwidth adaptation	The average downlink throughput of non-cell-center UEs increases. The benefits are related to the following factors: • A lower UE moving speed results in slower channel changes, less loss caused by the prolonged SRS frequency hopping period, and greater benefits of SRS bandwidth adaptation. A higher UE moving speed results in faster channel changes, greater loss caused by the prolonged SRS frequency hopping period, and less benefits of SRS bandwidth adaptation. • Greater benefit potential of SRS-based weights in NLOS scenarios results in greater benefits of SRS bandwidth adaptation. Less benefit potential of SRS-based weights in LOS scenarios results in less or even no benefits of SRS bandwidth adaptation. • SRS bandwidth adaptation provides greater benefits if there is a larger number of UEs meeting both of the following requirements: (1) "SRS-based digital weight" does not take effect before SRS frequency hopping is used. (2) "SRS-based digital weight" takes effect after SRS frequency hopping is used.

Function Name	Benefit
Precise power control	- After downlink PDSCH coverage enhancement is enabled, the average downlink throughput increases for UEs with this function in effect in the cell and the proportion^b of low rates^a decreases for the cell. This function is recommended in eMBB scenarios. It is recommended that PUCCH grouping adaptation enhancement for inter-FR CA be enabled (by selecting the PUCCH_GROUP_ADAPT_ENH_SW option of the NRDUCellCarrMgmt.InterFrCaEnhSw parameter) to achieve optimal PDSCH coverage, thereby further improving the average downlink throughput of these UEs. A higher proportion of inter-FR CA UEs using extremely low MCS indexes results in greater benefits. - After PDSCH DMRS power adaptation is enabled, the average downlink throughput increases for UEs with this function in effect in the cell and the proportion^b of low rates^a decreases for the cell. This function is recommended in eMBB scenarios. It is recommended that PUCCH grouping adaptation enhancement for inter-FR CA be enabled (by selecting the PUCCH_GROUP_ADAPT_ENH_SW option of the NRDUCellCarrMgmt.InterFrCaEnhSw parameter) to achieve optimal PDSCH coverage, thereby further improving the average downlink throughput of these UEs. A higher proportion of inter-FR CA UEs using extremely low MCS indexes results in greater benefits.
Downlink IBLER adaptation for CEUs	After downlink IBLER adaptation for CEUs is enabled, if the proportion of CEUs on the PDSCH is extremely large and their downlink signal quality fluctuates greatly, the average downlink throughput of such UEs may increase and the proportion ^b of low rates ^a in the cell decreases. This function is recommended in eMBB scenarios.
a: A rate is considered low if it is less than the result of "(Lower boundary of the throughput range corresponding to the non-zero counter with the largest index ID from N.Thp.DL.Samp.Index0 to N.Thp.DL.Samp.Index14) x 5%". For instance, if the value of N.Thp.DL.Samp.Index14 is 0, the counter with the highest index ID is N.Thp.DL.Samp.Index13 which corresponds to a throughput range of [512000 kbit/s, 1024000 kbit/s). In this case, a rate is considered low if it falls below 25600 kbit/s (25600 kbit/s = 512000 kbit/s x 5%). b: Proportion of low rates = Number of samples with low rates/Number of samples with all rates. If low rates refer to the rates below 25600 kbit/s and the throughput range associated with N.Thp.DL.Samp.Index6 is [25600 kbit/s, 30720 kbit/s), the number of samples with rates below 25600 kbit/s is the sum of counters ranging from N.Thp.DL.Samp.Index0 to N.Thp.DL.Samp.Index5. The proportion of low rates is calculated as follows: Proportion of low rates = (N.Thp.DL.Samp.Index0 + N.Thp.DL.Samp.Index1 + ... + N.Thp.DL.Samp.Index5)/(N.Thp.DL.Samp.Index0 + N.Thp.DL.Samp.Index1 + ... + N.Thp.DL.Samp.Index14).	

4.2.2 Impacts

The impacts between this function and other functions are described in [4.3.2.3 Function Impacts](#).

The following lists the network impacts of mmWave Super Coverage functions:

- After intelligent beam selection is enabled, the average uplink and downlink throughputs of UEs with this function not in effect may fluctuate due to possible changes in scheduling resources.
- After fixed SRS-based digital weight is enabled, the average downlink throughput of CEUs may decrease.
- After downlink PDSCH coverage enhancement is enabled, the proportion may increase, the downlink PRB usage may decrease, and the average downlink throughput may fluctuate for CEUs on the PDSCH.
- After downlink IBLER adaptation for CEUs is enabled, the downlink IBLER (**DL IBLER**) may increase. When the proportion of CEUs on the PDSCH is high, downlink data retransmissions increase due to UE distribution on different CSI-RS for BM beams or SSB beams. As a result, the average uplink or downlink UE throughput may fluctuate. When the proportion of CEUs is low, the average uplink and downlink UE throughputs may remain unchanged.

4.3 Requirements

4.3.1 Licenses

Feature ID	Feature Name	Model	Sales Unit
FOFD-101242	mmWave Super Coverage	NR0S00MWBM00	per Cell

NOTE

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists*.

4.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

4.3.2.1 Prerequisite Functions

- Intelligent beam selection

RAT	Function Name	Function Switch	Reference	Description
High-frequency TDD	Flexible Dense Beam	FLEXIBLE_DENSE_BEAM_SW option of the NRDUCellTrpMmwavBeam.BeamPerformanceSw parameter	<i>mmWave Beam Management (High-Frequency TDD)</i>	Intelligent beam selection can be enabled only if Flexible Dense Beam is enabled.
High-frequency TDD	mmWave super coverage	SUPER_COVERAGE_SW option of the NRDUCellChnCovAlgo.DL CoverageAlgoSwitch parameter	<i>mmWave Super Coverage (High-Frequency TDD)</i>	Intelligent beam selection can take effect only if mmWave super coverage is enabled.

- Beam tracking

RAT	Function Name	Function Switch	Reference	Description
High-frequency TDD	mmWave super coverage	SUPER_COVERAGE_SW option of the NRDUCellChnCovAlgo.DL CoverageAlgoSwitch parameter	<i>mmWave Super Coverage (High-Frequency TDD)</i>	Beam tracking can be enabled only if mmWave super coverage is enabled.

- SRS-based digital weight

RAT	Function Name	Function Switch	Reference	Description
High-frequency TDD	mmWave super coverage	SUPER_COVERAGE_SW option of the NRDUCellChnCovAlgo.DlCoverageAlgoSwitch parameter	<i>mmWave Super Coverage (High-Frequency TDD)</i>	- SRS- and PMI-based weight adaptation and SRS filtering-based downlink beamforming weight optimization can take effect only if mmWave super coverage is enabled. - Fixed SRS-based digital weight can take effect only if mmWave super coverage is enabled and SRS- and PMI-based weight adaptation is disabled.
High-frequency TDD	SRS- and PMI-based weight adaptation	DL_PMI_SRS_ADAPT_SW option of the NRDUCellAlgoSwitch.AdaptiveEdgeExpEnhSwitch parameter	<i>mmWave Super Coverage (High-Frequency TDD)</i>	- Fixed SRS-based digital weight can take effect only if mmWave super coverage is enabled and SRS- and PMI-based weight adaptation is disabled. - Downlink single-UE mobility precoding can be enabled only if fixed SRS-based digital weight or SRS- and PMI-based weight adaptation is enabled.
High-frequency TDD	Fixed SRS-based digital weight	NRDUCellPdsch.FixedWeightType set to SRS_WEIGHT	<i>mmWave Super Coverage (High-Frequency TDD)</i>	Downlink single-UE mobility precoding can be enabled only if fixed SRS-based digital weight or SRS- and PMI-based weight adaptation is enabled.
High-frequency TDD	SRS resource allocation for UEs performing downlink large-packet services	DL_BIG_PKT_SRS_RES_ALL_OC_SW option of the NRDUCellSrsSrsAlgoExtSwitch parameter	None	Downlink single-UE mobility precoding can be enabled only if SRS resource allocation for UEs performing downlink large-packet services is enabled.

- SRS bandwidth adaptation

RAT	Function Name	Function Switch	Reference	Description
High-frequency TDD	mmWave super coverage	SUPER_COVERAGE_SW option of the NRDUCellChnCovAlgo.DL CoverageAlgoSwitch parameter	<i>mmWave Super Coverage (High-Frequency TDD)</i>	SRS bandwidth adaptation can be enabled only if all the following conditions are met: - mmWave super coverage is enabled. - Fixed SRS-based digital weight or SRS- and PMI-based weight adaptation is enabled. - SRS resource allocation for UEs performing downlink large-packet services is enabled.
High-frequency TDD	SRS- and PMI-based weight adaptation	DL_PMI_SRS_ADAPT_SW option of the NRDUCellAlgoSwitch.AdaptiveEdgeExpEnhSwitch parameter	<i>mmWave Super Coverage (High-Frequency TDD)</i>	
High-frequency TDD	Fixed SRS-based digital weight	NRDUCellPdsch.FixedWeightType set to SRS_WEIGHT	<i>mmWave Super Coverage (High-Frequency TDD)</i>	
High-frequency TDD	SRS resource allocation for UEs performing downlink large-packet services	DL_BIG_PKT_SRS_RES_ALL_OC_SW option of the NRDUCellSrs.SrsAlgoExtSwitch parameter	None	

- Precise power control

RAT	Function Name	Function Switch	Reference	Description
High-frequency TDD	mmWave super coverage	SUPER_COVERAGE_SW option of the NRDUCellChnCovAlgo.DlCoverageAlgoSwitch parameter	<i>mmWave Super Coverage (High-Frequency TDD)</i>	- Downlink PDSCH coverage enhancement can take effect only if all of the following conditions are met: - mmWave super coverage is enabled. - The NRDUCellChnPwr.MaxPdschConvPwrOffset parameter is set to a value other than 0. - PUCCH grouping adaptation for inter-FR CA takes effect and the PCell and SCells use a single set of PUCCH-based feedback mechanisms. - PDSCH DMRS power adaptation can take effect only if both of the following conditions are met: - mmWave super coverage is enabled. - PUCCH grouping adaptation for inter-FR CA takes effect and the PCell and SCells use a single set of PUCCH-based feedback mechanisms.
High-frequency TDD	PUCCH grouping adaptation for inter-FR CA	PUCCH_GROUP_ADAPT_SW option of the NRDUCellCarrierMgmt.InterFrCaEnhSw parameter	<i>Carrier Aggregation</i>	

RAT	Function Name	Function Switch	Reference	Description
High-frequency TDD	Maximum Converged PDSCH Power	NRDUCellChnPwr.MaxPdschConvPwrOffset	<i>Power Control</i>	Downlink PDSCH coverage enhancement can take effect only if all of the following conditions are met: • mmWave super coverage is enabled. • The NRDUCellChnPwr.MaxPdschConvPwrOffset parameter is set to a value other than 0. • PUCCH grouping adaptation for inter-FR CA takes effect and the PCell and SCells use a single set of PUCCH-based feedback mechanisms.

- Downlink IBLER adaptation for CEUs

RAT	Function Name	Function Switch	Reference	Description
High-frequency TDD	mmWave super coverage	SUPER_COVERAGE_SW option of the NRDUCellChnCovAlgo.DL CoverageAlgorithmSwitch parameter	<i>mmWave Super Coverage (High-Frequency TDD)</i>	Downlink IBLER adaptation for CEUs can take effect only if mmWave super coverage is enabled.

4.3.2.2 Mutually Exclusive Functions

- Intelligent beam selection

RAT	Function Name	Function Switch	Reference	Description
High-frequency TDD	mmWave 3D coverage pattern	NRDUCellTrpMmwavBeam.CoverageScenario	<i>mmWave Beam Management (High-Frequency TDD)</i>	In the single-beam scenario of mmWave 3D coverage pattern (with the NRDUCellTrpMmwavBeam.CoverageScenario parameter set to SCENARIO_103(Single Beam Scenario)), intelligent beam selection cannot be enabled.
High-frequency TDD	Hyper Cell	NRDUCell.NrDuCellNetworkingMode set to HYPER_CELL	None	Intelligent beam selection cannot be enabled when the NRDUCell.NrDuCellNetworkingMode parameter is set to HYPER_CELL .
High-frequency TDD	Combined cell	NRDUCell.NrDuCellNetworkingMode set to HYPER_CELL_COMBINE_MODE	None	Intelligent beam selection cannot be enabled when the NRDUCell.NrDuCellNetworkingMode parameter is set to HYPER_CELL_COMBINE_MODE .

- Beam tracking

RAT	Function Name	Function Switch	Reference	Description
High-frequency TDD	Hyper Cell	NRDUCell.NrDuCellNetworkingMode set to HYPER_CELL	None	Beam tracking cannot be enabled when the NRDUCell.NrDuCellNetworkingMode parameter is set to HYPER_CELL .
High-frequency TDD	Combined cell	NRDUCell.NrDuCellNetworkingMode set to HYPER_CELL_COMBINE_MODE	None	Beam tracking cannot be enabled when the NRDUCell.NrDuCellNetworkingMode parameter is set to HYPER_CELL_COMBINE_MODE .

- SRS-based digital weight

RAT	Function Name	Function Switch	Reference	Description
High-frequency TDD	Hyper Cell	NRDUCell.NrDuCellNetworkingMode set to HYPER_CELL	None	Fixed SRS-based digital weight and SRS- and PMI-based weight adaptation cannot be enabled when the NRDUCell.NrDuCellNetworkingMode parameter is set to HYPER_CELL .

- SRS bandwidth adaptation

RAT	Function Name	Function Switch	Reference	Description
High-frequency TDD	PUCCH resource reduction for heavy-traffic UEs	NRDUCellPucch.PucchResourceRednMthd set to MASS_FLOW	<i>Channel Management</i>	SRS bandwidth adaptation cannot be enabled when the NRDUCellPucch.PucchResourceRednMthd parameter is set to MASS_FLOW .
High-frequency TDD	Increased downlink symbol proportion in self-contained slots	S_SLOT_DL_SYM_RATIO_INC_SW option of the NRDUCellDischRes.DischResAlgoSw parameter	None	SRS bandwidth adaptation cannot be enabled together with increased downlink symbol proportion in self-contained slots.
High-frequency TDD	AoA- and UTDOA-based hybrid positioning	AOA_UTDOA_HYBRID_LOCATION_SW option of the NRCelAlgoSwitch.LcsSwitch parameter	None	SRS bandwidth adaptation cannot be enabled together with AoA- and UTDOA-based hybrid positioning.

- Precise power control
None
- Downlink IBLER adaptation for CEUs
None

4.3.2.3 Function Impacts

- Intelligent beam selection

RAT	Function Name	Function Switch	Reference	Description
High-frequency TDD	PUCCH grouping adaptation for inter-FR CA	PUCCH_GROUP_ADAPT_SW option of the NRDUCellCarrierMgmt.InterFrCaEnhSw parameter	<i>Carrier Aggregation</i>	Intelligent beam selection does not take effect when PUCCH grouping adaptation for inter-FR CA is in effect, in which case a single set of PUCCH-based feedback mechanisms is used in the PCell and SCells of a UE in the inter-FR CA state.
High-frequency TDD	RF channel intelligent shutdown	RF_SHUTDOWN_SW option of the NRDUCellAlgorithmSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	Intelligent beam selection does not take effect when RF channel intelligent shutdown has shut down RF channels to have 2T left.

- Beam tracking
None
- SRS-based digital weight
None
- SRS bandwidth adaptation
None
- Precise power control

RAT	Function Name	Function Switch	Reference	Description
High-frequency TDD	PDCCH rate matching	PDCCH_RATE_MATCH_SWITCH option of the NRDUCellPdsch.RateMatchSwitch parameter PDCCH_PATTERN_RATE_MATCH_SWITCH option of the NRDUCellPdcch.PdcchAlgoSwitch parameter	<i>Downlink Scheduling</i>	Rate-matching-pattern-configuration-based PDCCH rate matching and rate-matching-pattern-configuration-free PDCCH rate matching do not take effect in a TTI when downlink PDSCH coverage enhancement is in effect and the transmit PSD of the base station is adjusted.

- Downlink IBLER adaptation for CEUs
None

4.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

5900 series base stations (macro base stations). These base stations must be equipped with BBUs other than Book BBUs when intelligent beam selection, beam tracking, SRS-based digital weight, or SRS bandwidth adaptation is used.

Boards

All NR-capable main control boards and NR TDD mmWave baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

HAAU5235w, HAAU5323, HAAU5233w

4.3.4 Cells

- Intelligent beam selection
 - Intelligent beam selection takes effect only if the switches for mmWave super coverage and intelligent beam selection are both turned on for all activated cells in the same sector.
 - The **NRDUCell.SlotAssignment** parameter cannot be set to **2_3_DSUUU**.
- Beam tracking

Beam tracking takes effect only if the beam tracking switch is turned on for all activated cells in the same sector.
- SRS-based digital weight
 - Fixed SRS-based digital weight and SRS- and PMI-based weight adaptation do not take effect when the sector-level downlink bandwidth (sum of **NRDUCell.DlBandwidth** for all cells in a sector) exceeds 400 MHz in FWA and eMBB scenarios.
 - SRS- and PMI-based weight adaptation cannot be enabled if the **NRDUCell.SlotAssignment** parameter is set to **2_3_DSUUU**.
 - SRS- and PMI-based weight adaptation cannot be enabled if the **NRDUCellTRP.TxRxMode** parameter is set to **2T2R**.
- SRS bandwidth adaptation

SRS bandwidth adaptation can take effect only if all activated cells in the same sector meet the following conditions:

 - The SRS bandwidth adaptation switch is turned on for all activated cells.
 - Fixed SRS-based digital weight is configured for all activated cells, or SRS- and PMI-based weight adaptation is configured for all activated cells.

4.3.5 Others

- Intelligent beam selection

Requirements on UEs:

 - The value of **maxNumberCSI-RS-Resource** is greater than or equal to n32.
 - The value of **maxNumberNonGroupBeamReporting** is n4.

For details about definitions of **maxNumberCSI-RS-Resource** and **maxNumberNonGroupBeamReporting**, see section 4.2.7.2 "BandNR parameters" in 3GPP TS 38.306 V17.4.0.
- Other functions

None

4.4 Operation and Maintenance

4.4.1 Data Configuration

4.4.1.1 Data Preparation

Table 4-3 describes the parameters used for function activation.

Table 4-3 Parameters used for activation

Parameter Name	Parameter ID	Option	Setting Notes
DL Coverage Algorithm Switch	NRDUCellChnCovAlgo.DLCoverageAlgoSwitch	SUPER_COVERAGE_SW	It is recommended that this option be selected.
Beam Optimization Algorithm Switch	NRDUCellBeamAlgo.BeamOptAlgoSwitch	INTELLIGENT_BEAM_SELECTION_SW	It is recommended that this option be selected.
Beam Optimization Algorithm Switch	NRDUCellBeamAlgo.BeamOptAlgoSwitch	BEAM_TRACKING_SW	It is recommended that this option be selected in eMBB scenarios.
Adaptive Edge Experience Enhance Switch	NRDUCellAlgoSwitch.AdaptiveEdgeExpEnhSwitch	DL_PMI_SRS_ADAPT_SW	It is recommended that this option be selected.
Adaptive Edge Experience Enhance Switch	NRDUCellAlgoSwitch.AdaptiveEdgeExpEnhSwitch	DL_BF_WEIGHTFILTER_OPT_SW	It is recommended that this option be selected in FWA scenarios.
DL SU-MIMO Mobility Precoding Mode	NRDUCellPdschPre-code.DLSuMimoMobPrecodingMode	N/A	It is recommended that this parameter be set to ROBUST_PHASEIII in eMBB scenarios.
Fixed Weight Type	NRDUCellPdsch.FixedWeightType	N/A	It is recommended that this parameter be set to SRS_WEIGHT .
SRS Algorithm Extension Switch	NRDUCellSrs.SrsAlgoExtSwitch	SRS_BANDWIDTH_ADAPT_SW	It is recommended that this option be selected.
Channel Power Algorithm Switch	NRDUCellChnPwr.ChnPwrAlgoSwitch	PDSCH_COV_ENHANCE_SW	It is recommended that this option be selected in eMBB scenarios.

Parameter Name	Parameter ID	Option	Setting Notes
Channel Power Algorithm Switch	NRDUCellChnPwr.ChnPwrAlgoSwitch	DMRS_PWR_ADAPT_SW	It is recommended that this option be selected in eMBB scenarios.
DL MCS Selection Algorithm Switch	NRDUCellDlAmc.DlMcsSelAlgoSw	DL_EDGE_INTER_ADAPT_SW	It is recommended that this option be selected in eMBB scenarios.
Maximum Converged PDSCH Power	NRDUCellChnPwr.MaxPdschConvPwrOffset	N/A	It is recommended that this parameter be set to a value greater than or equal to 3 if downlink PDSCH coverage enhancement needs to be enabled.

4.4.1.2 Using MML Commands

Before using MML commands, refer to [4.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

```
//Turning on the mmWave super coverage switch
MOD NRDUCELLCHNCOVALGO: NrDuCellId=0, DlCoverageAlgoSwitch=SUPER_COVERAGE_SW-1;
//Enabling intelligent beam selection
MOD NRDUCELLBEAMALGO: NrDuCellId=0, BeamOptAlgoSwitch=INTELLIGENT_BEAM_SELECTION_SW-1;
//Enabling beam tracking
MOD NRDUCELLBEAMALGO: NrDuCellId=0, BeamOptAlgoSwitch=BEAM_TRACKING_SW-1;
//Enabling SRS- and PMI-based weight adaptation
MOD NRDUCELLALGOSWITCH: NrDuCellId=0, AdaptiveEdgeExpEnhSwitch=DL_PMI_SRS_ADAPT_SW-1;
//Enabling fixed SRS-based digital weight
MOD NRDUCELLALGOSWITCH: NrDuCellId=0, AdaptiveEdgeExpEnhSwitch=DL_PMI_SRS_ADAPT_SW-0;
MOD NRDUCELLPDSCH: NrDuCellId=0, FixedWeightType=SRS_WEIGHT;
//Enabling downlink single-UE mobility precoding
MOD NRDUCELLPDSCHPRECODE: NrDuCellId=0, DISuMimoMobPrecodeMode=ROBUST_PHASEIII;
//Enabling SRS filtering-based downlink beamforming weight optimization
MOD NRDUCELLALGOSWITCH: NrDuCellId=0,
AdaptiveEdgeExpEnhSwitch=DL_BF_WEIGHTFILTER_OPT_SW-1;
//Enabling SRS bandwidth adaptation
MOD NRDUCELLSRS: NrDuCellId=0, SrsAlgoExtSwitch=SRS_BANDWIDTH_ADAPT_SW-1;
//Enabling downlink PDSCH coverage enhancement
```

```
MOD NRDUCELLCHNPWR: NrDuCellId=0,ChnPwrAlgoSwitch=PDSCH_COV_ENHANCE_SW-1;  
//Configuring Maximum Converged PDSCH Power (using the value of 3 as an example)  
MOD NRDUCELLCHNPWR: NrDuCellId=0, MaxPdschConvPwrOffset=3;  
//Enabling PDSCH DMRS power adaptation  
MOD NRDUCELLCHNPWR: NrDuCellId=0,ChnPwrAlgoSwitch=DMRS_PWR_ADAPT_SW-1;  
//Enabling downlink IBLER adaptation for CEUs  
MOD NRDUCELLDLAMC: NrDuCellId=0,DIMcsSelAlgoSw=DL_EDGE_IBLER_ADAPT_SW-1;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//Disabling intelligent beam selection  
MOD NRDUCELLBEAMALGO: NrDuCellId=0, BeamOptAlgoSwitch=INTELLIGENT_BEAM_SELECTION_SW-0;  
//Disabling beam tracking  
MOD NRDUCELLBEAMALGO: NrDuCellId=0, BeamOptAlgoSwitch=BEAM_TRACKING_SW-0;  
//Disabling downlink single-UE mobility precoding  
MOD NRDUCELLPDSCHPRECODE: NrDuCellId=0, DISuMimoMobPrecodeMode=NOT_CFG;  
//Disabling SRS filtering-based downlink beamforming weight optimization  
MOD NRDUCELLALGOSWITCH: NrDuCellId=0,  
AdaptiveEdgeExpEnhSwitch=DL_BF_WEIGHTFILTER_OPT_SW-0;  
//Disabling SRS- and PMI-based weight adaptation  
MOD NRDUCELLALGOSWITCH: NrDuCellId=0, AdaptiveEdgeExpEnhSwitch=DL_PMI_SRS_ADAPT_SW-0;  
//Disabling fixed SRS-based digital weight  
MOD NRDUCELLPDSCH: NrDuCellId=0, FixedWeightType=PMI_WEIGHT;  
//Disabling SRS bandwidth adaptation  
MOD NRDUCELLSRS: NrDuCellId=0, SrsAlgoExtSwitch=SRS_BANDWIDTH_ADAPT_SW-0;  
//Disabling downlink PDSCH coverage enhancement  
MOD NRDUCELLCHNPWR: NrDuCellId=0,ChnPwrAlgoSwitch=PDSCH_COV_ENHANCE_SW-0;  
//Disabling PDSCH DMRS power adaptation  
MOD NRDUCELLCHNPWR: NrDuCellId=0,ChnPwrAlgoSwitch=DMRS_PWR_ADAPT_SW-0;  
//Disabling downlink IBLER adaptation for CEUs  
MOD NRDUCELLDLAMC: NrDuCellId=0,DIMcsSelAlgoSw=DL_EDGE_IBLER_ADAPT_SW-0;  
//Turning off the mmWave super coverage switch  
MOD NRDUCELLCHNCOVALGO: NrDuCellId=0, DICoverageAlgoSwitch=SUPER_COVERAGE_SW-0;
```

4.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

4.4.2 Activation Verification

- Intelligent beam selection
After this function is enabled, run the **DSP NRDUCELLBMAIMODEL** command with the **NRDUCellBmAiModel.AiCase** parameter set to **INTELLIGENT_BEAM_SELECTION**. Check the value of **AI Model Status**. If the value is **Building**, **In Use**, or **Updating**, intelligent beam selection has taken effect.
- Beam tracking
After this function is enabled, run the **DSP NRDUCELLBMAIMODEL** command with the **NRDUCellBmAiModel.AiCase** parameter set to **BEAM_TRACKING**. Check the value of **AI Model Status**. If the value is **Building** or **In Use**, beam tracking has taken effect.

- SRS-based digital weight
After being enabled, this function has taken effect if the value of **User Downlink Average Throughput (DU)** increases.
- SRS bandwidth adaptation
 - a. On the MAE-Access, choose **Monitor > Signaling Trace > Signaling Trace Management**.
 - b. In the navigation tree on the left of the **Signaling Trace Management** tab page, choose **NR > Application Layer > Uu Interface Trace**.
 - c. Observe the RRC_CONN_RECFG message to check whether this function takes effect. If the value of the *b-SRS* or *b-hop* IE in *srs-Config* under *bwp-Dedicated* in the message is not **0x0** and the value of *resourceType* is **periodic**, SRS bandwidth adaptation has taken effect.
- Precise power control
After being enabled, this function has taken effect if the average MCS index of CEUs increases.
$$\text{Average MCS index of CEUs} = (0 \times \text{N.ChMeas.PDSCH.MCS.0} + 1 \times \text{N.ChMeas.PDSCH.MCS.1} + 2 \times \text{N.ChMeas.PDSCH.MCS.2} + 3 \times \text{N.ChMeas.PDSCH.MCS.3}) / (\text{N.ChMeas.PDSCH.MCS.0} + \text{N.ChMeas.PDSCH.MCS.1} + \text{N.ChMeas.PDSCH.MCS.2} + \text{N.ChMeas.PDSCH.MCS.3})$$
- Downlink IBLER adaptation for CEUs
After being enabled, this function has taken effect if the downlink IBLER (**DL IBLER**) increases. Specifically:
$$\text{DL IBLER} = (\text{N.DL.SCH.QPSK.ErrTB.Ibler} + \text{N.DL.SCH.16QAM.ErrTB.Ibler} + \text{N.DL.SCH.64QAM.ErrTB.Ibler} + \text{N.DL.SCH.256QAM.ErrTB.Ibler}) / (\text{N.DL.SCH.QPSK.TB} + \text{N.DL.SCH.16QAM.TB} + \text{N.DL.SCH.64QAM.TB} + \text{N.DL.SCH.256QAM.TB}) \times 100\%$$

4.4.3 Network Monitoring

The indicators listed in [4.2 Network Analysis](#) can be used to monitor the benefits and impacts of the functions related to this feature.

5 Parameters

The following hyperlinked EXCEL files of parameter reference match the software version with which this document is released.

- Node Parameter Reference: contains device and transport parameters.
- Node Used Reserved Parameter List: contains the reserved parameters that are in use and those that have been disused.
- gNodeBFunction Parameter Reference: contains all parameters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- 3900 & 5900 Series Base Station gNodeBFunction Used Reserved Parameter List: contains the reserved parameters that are in use and the reserved parameters that fall into disuse in the current R version.

NOTE

You can find the EXCEL files of parameter reference and used reserved parameter list for the software version used on the live network from the "Man-Machine Interface Reference" node in the *3900 & 5900 Series Base Station Product Documentation* delivered with that version.

FAQ 1: How do I find the parameters related to a certain feature from parameter reference?

- Step 1** Open the EXCEL file of parameter reference.
- Step 2** On the **Parameter List** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID, for example, FBFD-010011.
- Step 3** Click **OK**. All parameters related to the feature are displayed.

----End

FAQ 2: How do I find the information about a certain reserved parameter from the used reserved parameter list?

- Step 1** Open the EXCEL file of the used reserved parameter list.

Step 2 On the **Used Reserved Parameter List** sheet, use the **MO**, **Parameter ID**, and **BIT** columns to locate the reserved parameter. View its information, including the meaning, values, and impacts.

----End

6 Counters

The following hyperlinked EXCEL files of performance counter reference match the software version with which this document is released.

- Node Performance Counter Summary: contains device and transport counters.
- gNodeBFunction Performance Counter Summary: contains all counters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- gNodeBFunction Used Reserved Counter List: contains the reserved counters that are in use and those that have been disused.

NOTE

You can find the EXCEL files of performance counter reference and used reserved counter list for the software version used on the live network from the "Man-Machine Interface Reference" node in the *3900 & 5900 Series Base Station Product Documentation* delivered with that version.

FAQ: How do I find the counters related to a certain feature from performance counter reference?

- Step 1** Open the EXCEL file of performance counter reference.
- Step 2** On the **Counter Summary(En)** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID, for example, FBFD-010011.
- Step 3** Click **OK**. All counters related to the feature are displayed.

----End

7 Glossary

For the acronyms, abbreviations, terms, and definitions, see *Glossary*.

8 Reference Documents

- 3GPP TS 38.104: "NR; Base Station (BS) radio transmission and reception"
- 3GPP TS 38.214: "NR; Physical layer procedures for data"
- 3GPP TS 38.306: "NR; User Equipment (UE) radio access capabilities"
- *mmWave Beam Management (High-Frequency TDD)*
- *MIMO (TDD)*
- *FWA*
- *Carrier Aggregation*
- *Downlink Scheduling*
- *Power Control*
- *Channel Management*
- *Energy Conservation and Emission Reduction*
- *License Control Item Lists*
- *Technical Specifications in 3900 & 5900 Series Base Station Product Documentation*